Revision 2021 Engineering Handbook

Strength of Materials

A professional diploma engineering textbook for stress, strain, shear force, bending moment, simple bending, deflection of beams, columns, shafts, springs and thin cylindrical shells.

Course Code
3021

Semester
3/3/3/4

Credits
3/3/3/No Credit

Periods
45

Learning Flow

This subject is the bridge between Engineering Mechanics, Machine Design, structures, machine tools and maintenance.

Course Overview

Official information

Program	Mechanical / Tool and Die / Manufacturing Technology / Wood and Paper Technology
Course title	Strength of Materials
Category	Program Core
L:T:P	2:1:0

Objectives

1. Understand tensile, compressive and shear stresses.
2. Show shear force and bending moment diagrams.
3. Explain simple bending and deflection of beams.
4. Discuss torsion, springs and thin cylindrical shells.

Prerequisites

- Basic Mathematics: Mathematics I and II.
- Basic Physics: Applied Physics I and II.
- Mechanics basics: Engineering Mechanics.

Suggested practical experiments: Not specified in supplied syllabus/source.

> Malayalam support: Formula □□□□□□□ □□□□□□□□□ □□□□. Load path, diagram, sign convention, unit conversion, final unit □□□□□□□ correct □□□□□□□.

Redesigned Syllabus Roadmap

CO1 / 11 h

Simple stresses and strains

Draw stress-strain curve. Calculate stress, strain, elongation, lateral and volumetric strain, thermal stress.

CO2 / 10 h

SFD and BMD

Types of beams and loads. Draw SFD/BMD for cantilever and simply supported beams with point load and UDL.

CO3 / 11 h

Bending, deflection and columns

Bending equation, deflection formulae, Euler formula, Rankine formula, effective length and slenderness ratio.

CO4 / 13 h

Shafts, springs and cylinders

Torsion equation, polar M.I., helical spring deflection/stiffness, hoop and longitudinal stress.

Module 1: Simple Stresses, Strains and Thermal Stress

Stress is internal resistance per unit area. Strain is deformation per original dimension. Tensile, compressive and shear forces are the basic force types. Mechanical properties such as elasticity, plasticity, ductility, brittleness and toughness explain how materials behave under load.

Important drawings: stress-strain diagram for mild steel and cast iron, uniform bar, composite bar, restrained bar under temperature change.

Key exam method: write definition, unit, formula, substitute in N and mm, then state final answer and nature of stress.

Module 2: Shear Force and Bending Moment Diagrams

A beam carries transverse loads. Shear force at a section is the algebraic sum of vertical forces on one side. Bending moment is the algebraic sum of moments of forces on one side.

Point load causes a sudden jump in SFD. UDL causes sloping SFD and parabolic BMD. Maximum BM occurs where SF is zero or changes sign.

Drawing sequence: beam sketch, support reactions, SF values, SFD, BM values, BMD, maximum value marking.

Module 3: Simple Bending, Deflection, Columns and Struts

The bending equation is $M/I = \sigma/y = E/R$. Neutral axis is the layer where bending stress is zero. Section modulus $Z = I/y_{max}$, and maximum stress is $\sigma = M/Z$.

Deflection is important for serviceability. Columns fail by buckling when slender. Effective length depends on end condition. Euler formula is used for long columns and Rankine formula is practical for intermediate columns.

Module 4: Torsion in Shafts, Helical Springs and Thin Cylindrical Shells

Torsion equation: $T/J = \tau/R = G\theta/L$. Solid shaft polar M.I. is $\pi d^4/32$. Hollow shafts are efficient because material away from the centre resists torsion better.

Close-coiled helical spring deflection: $\delta = 8WD^3n/Gd^4$. Spring stiffness $k = W/\delta$. Thin cylinder hoop stress $\sigma_h = pD/2t$ and longitudinal stress $\sigma_l = pD/4t$.

Formula Bank

Stress: $\sigma = P/A$. Unit N/mm².

Strain: $\varepsilon = \delta L/L$. No unit.

Elongation: $\delta L = PL/AE$.

Thermal stress: $\sigma = E\cdot\alpha\cdot\Delta T$.

Bending: $M/I = \sigma/y = E/R$.

Euler load: $P_{cr} = \pi^2 EI/Le^2$.

Torsion: $T/J = \tau/R = G\theta/L$.

Spring deflection: $\delta = 8WD^3n/Gd^4$.

Hoop stress: $\sigma_h = pD/2t$.

Solved Examples

1. Axial bar: P = 30 kN, A = 300 mm². Stress σ = 30000/300 = 100 N/mm². If E = 200000 N/mm², strain ε = 0.0005 and elongation δL over 1200 mm is 0.6 mm.
2. Thermal stress: E = 200000 N/mm², $\alpha = 12\times10^{-6}$/°C, ΔT = 50°C. Stress σ = 120 N/mm² compressive if expansion is prevented.
3. UDL beam: span=6 m, w=4 kN/m. Reactions=12 kN each, maximum BM = $wL^2/8$ = 18 kN·m.
4. Shaft: P=8 kW, N=300 rpm. Torque T = 9550P/N = 254.67 N·m. Use $T = \pi\cdot\tau\cdot d^3/16$ for diameter.

Practice and Expected Questions

1. Draw stress-strain diagram for mild steel.
2. Explain factor of safety and elastic constants.
3. Draw SFD and BMD for cantilever with point load.
4. Draw SFD and BMD for simply supported beam with UDL.
5. State and explain bending equation.
6. Explain Euler and Rankine column formulae.
7. Compare solid and hollow shafts.
8. Solve helical spring deflection and thin cylinder thickness problems.

Fresh Model Question Paper

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Time: 3 Hours | Maximum Marks: 100
Part A: Define stress, strain, Hooke's law, factor of safety, shear force,
bending moment, neutral axis, Euler load and spring stiffness.
Part B: Explain stress-strain diagram, SFD/BMD procedure, bending equation,
column end conditions, shaft comparison and cylinder stresses.
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Part C: Solve numerical problems on axial deformation, thermal stress, SFD/BMD,
bending stress, beam deflection, Euler buckling load, shaft diameter, spring
deflection and cylinder thickness.
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Complete Model Answer Guide

Stress/strain answers must include definition, unit and formula. SFD/BMD answers must include reactions, values and diagrams. Bending answers must include M/I = σ/y = E/R and variable meaning. Thin cylinder answers must compare hoop and longitudinal stress and use the governing stress.

References

1. R. K. Bansal - Strength of Materials.
2. R. S. Khurmi - Strength of Materials.
3. S. S. Bhavikatti - Strength of Materials.
4. Ramamrutham - Strength of Materials.
5. T. D. Gunneswara Rao - Strength of Materials.